

Calculus I Lesson 25B

Alternative definition of a function $f(x)$ at c : $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$

1) Find the minimum function value of $f(x) = |x| + 2 = \begin{cases} x + 2 & \text{if } x \geq 0 \\ -x + 2 & \text{if } x < 0 \end{cases}$

a) $f(x) = |x| + 2$ has a minimum at the point ____?

b) Find $f'(x)$ at $x = 0$:

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x + 2 - 2}{x - 0} = \underline{\quad ? \quad} \quad (\text{Note: } x \rightarrow 0 \text{ from the left, } f(x) = -x + 2)$$

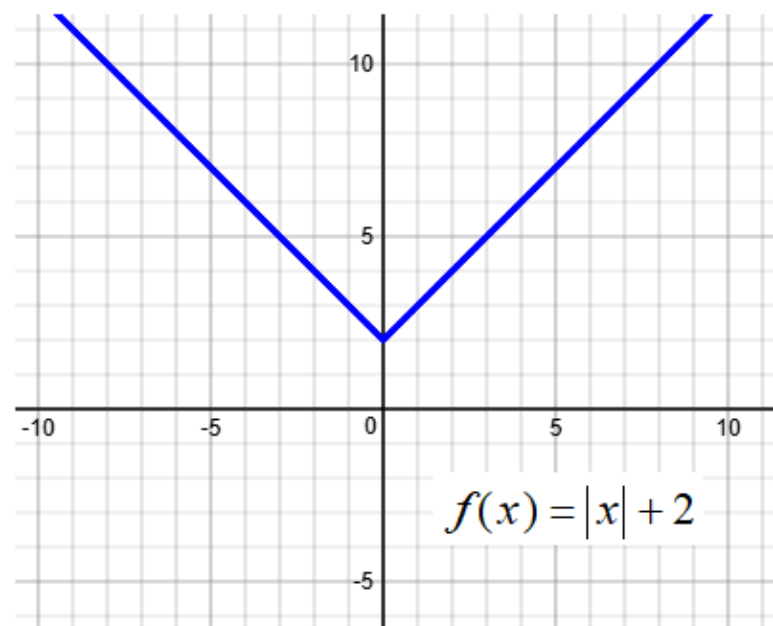
$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x + 2 - 2}{x - 0} = \underline{\quad ? \quad} \quad (\text{Note: } x \rightarrow 0 \text{ from the right, } f(x) = x + 2)$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \underline{\quad ? \quad}$$

c) Is $f'(0)$ defined?

d) Is the function $f(x) = |x| + 2$ differentiable at $x = 0$?

e) A critical number of $f(x)$ is ____?



2) Find the minimum function value of $f(x) = |x - 3| = \begin{cases} x - 3 & \text{if } x - 3 \geq 0 \\ -(x - 3) & \text{if } x - 3 < 0 \end{cases}$

a) $f(x) = |x - 3|$ has a minimum at the point _____?

b) Find $f'(x)$ at $x = 3$:

$$\lim_{x \rightarrow 3^-} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3^-} \frac{-(x - 3) - 0}{x - 3} = \underline{\quad ? \quad} \quad (\text{Note: } x \rightarrow 3 \text{ from the left, } f(x) = -(x - 3))$$

$$\lim_{x \rightarrow 3^+} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3^+} \frac{x - 3 - 0}{x - 3} = \underline{\quad ? \quad} \quad (\text{Note: } x \rightarrow 3 \text{ from the right, } f(x) = x - 3)$$

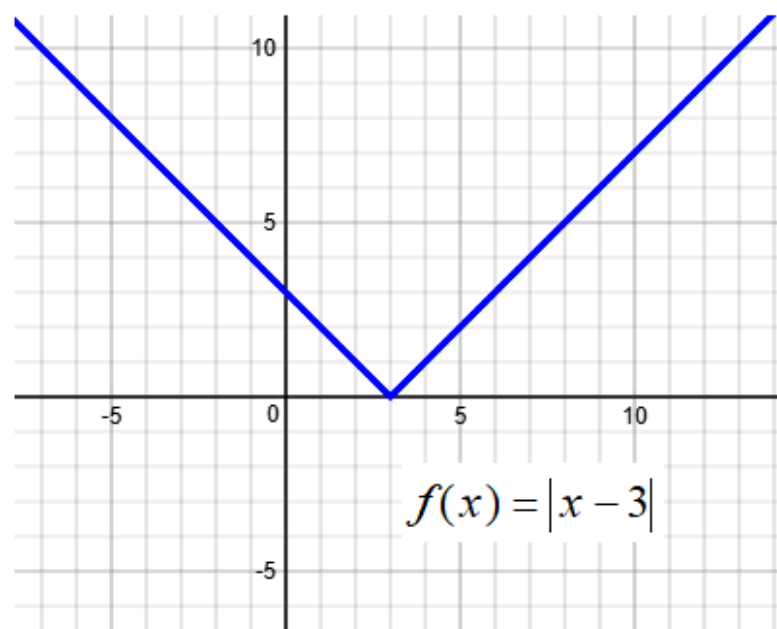
$$f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \underline{\quad ? \quad}$$

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c) Is $f'(3)$ defined?

d) Is the function $f(x) = |x - 3|$ differentiable at $x = 3$?

e) A critical number of $f(x)$ is _____?



3) Find the maximum function value of

$$f(x) = 4 - |x + 1| = \begin{cases} 4 - (x + 1) = 3 - x & \text{if } x + 1 \geq 0 \\ 4 - -(x + 1) = 5 + x & \text{if } x + 1 < 0 \end{cases}$$

a) $f(x) = 4 - |x + 1|$ has a maximum at the point _____?

b) Find $f'(x)$ at $x = -1$:

$$\lim_{x \rightarrow -1^-} \frac{f(x) - f(-1)}{x - (-1)} = \lim_{x \rightarrow -1^-} \frac{5 + x - 4}{x + 1} = \underline{\quad ? \quad} \quad [\text{Note: } x \rightarrow -1 \text{ from the left, } f(x) = 5 + x]$$

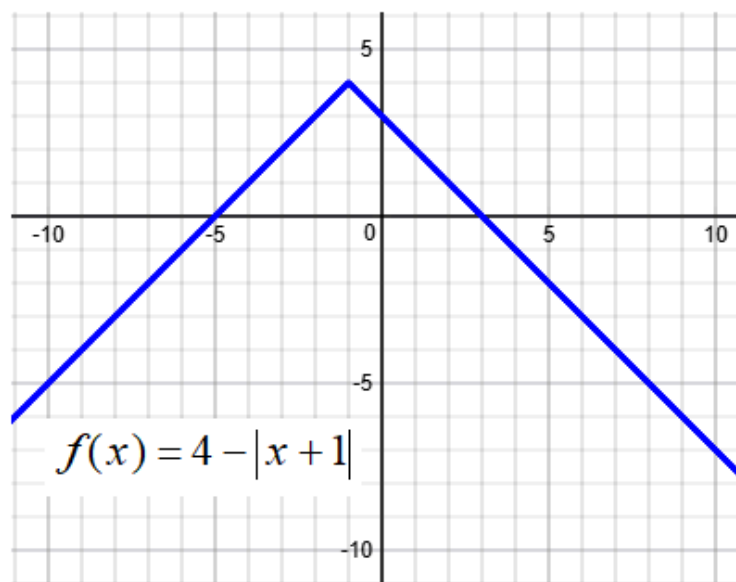
$$\lim_{x \rightarrow -1^+} \frac{f(x) - f(-1)}{x - (-1)} = \lim_{x \rightarrow -1^+} \frac{3 - x - 4}{x + 1} = \underline{\quad ? \quad} \quad [\text{Note: } x \rightarrow -1 \text{ from the right, } f(x) = x - 3]$$

$$f'(-1) = \lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)} = \underline{\quad ? \quad}$$

c) Is $f'(-1)$ defined?

d) Is the function $f(x) = 4 - |x + 1|$ differentiable at $x = -1$?

e) A critical number of $f(x)$ is _____?



4) Find the maximum function value of

$$f(x) = 2 - |3x + 6| = \begin{cases} 2 - (3x + 6) = -4 - 3x & \text{if } 3x + 6 \geq 0 \\ 2 - (-3x - 6) = 8 + 3x & \text{if } 3x + 6 < 0 \end{cases}$$

a) $f(x) = 2 - |3x + 6|$ has a maximum at the point ____?

b) Find $f'(x)$ at $x = -2$:

$$\lim_{x \rightarrow -2^-} \frac{f(x) - f(-2)}{x - (-2)} = \lim_{x \rightarrow -2^-} \frac{8 + 3x - 2}{x + 2} = \underline{\quad? \quad} \quad [\text{Note: } x \rightarrow -2 \text{ from the left, } f(x) = 8 + 3x]$$

$$\lim_{x \rightarrow -2^+} \frac{f(x) - f(-2)}{x - (-2)} = \lim_{x \rightarrow -2^+} \frac{-4 - 3x - 2}{x + 2} = \underline{\quad? \quad} \quad [\text{Note: } x \rightarrow -2 \text{ from the right, } f(x) = -4 - 3x]$$

$$f'(-2) = \lim_{x \rightarrow -2} \frac{f(x) - f(-2)}{x - (-2)} = \underline{\quad? \quad}$$

c) Is $f'(-2)$ defined?

d) Is the function $f(x) = 2 - |3x + 6|$ differentiable at $x = -2$?

e) A critical number of $f(x)$ is ____?

